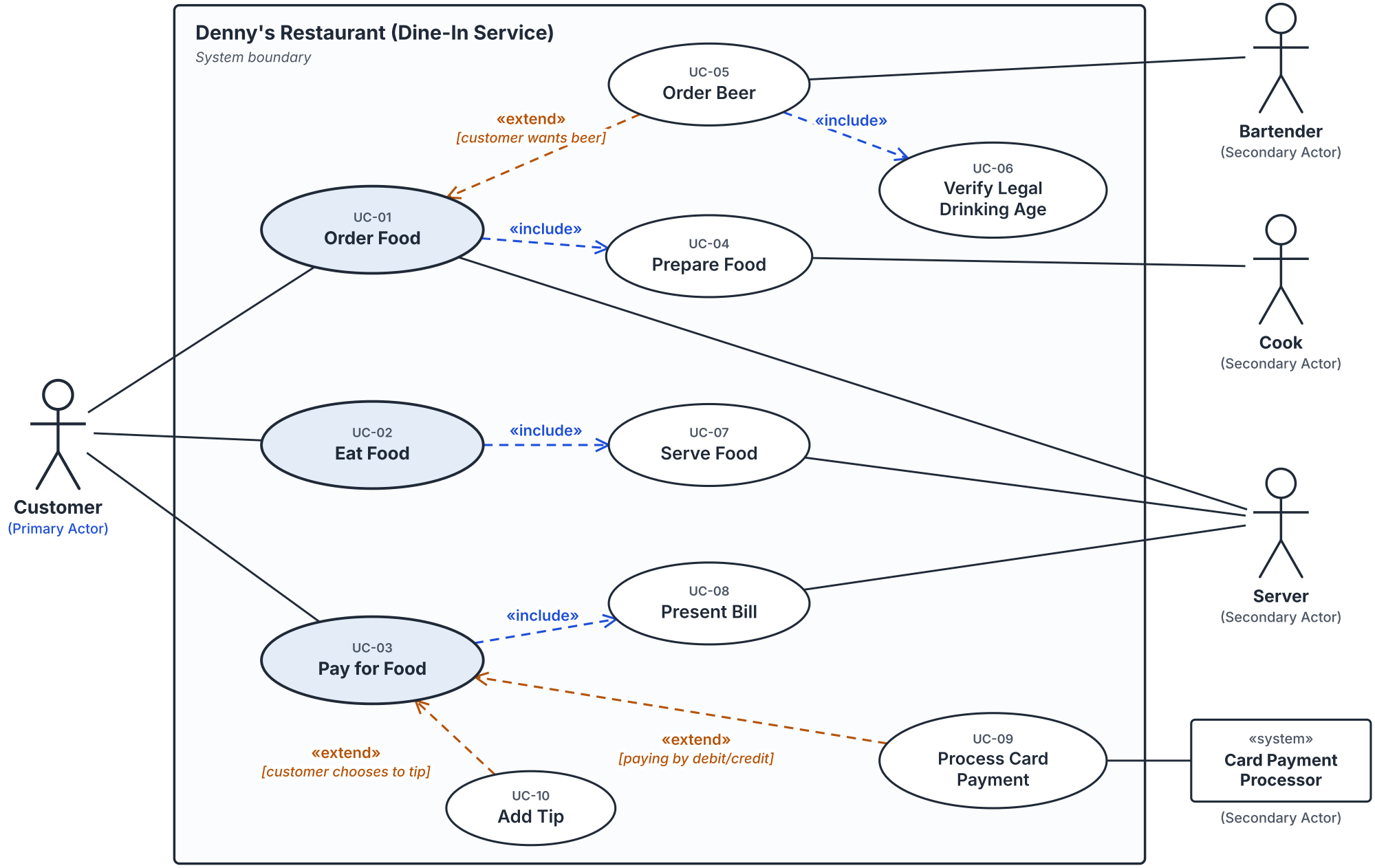
Use Case Diagram: Dining at Denny's Restaurant

MGMT6124 Methods for Business Analysis | Module 8: Process Modelling 1 | Adekunle Adeniji | November 3, 2026



LEGEND

- Association: the actor takes part in the use case
- Shaded oval: base use case (a customer goal in the scenario)
- «include» mandatory, always performed (points to the included use case)
- «extend» optional, only when the [condition] is true (points to the base)

Included and extending use cases run inside their base use case, so actors linked to the base (e.g. the Customer and Server) take part in them without a direct line.

Use Case Diagram: Supporting Notes

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1. Scenario and scope

A customer at a Denny's restaurant orders food, eats it and pays for it, and may also order beer with dinner. The system modelled is the restaurant's **dine-in service**. The scenario starts once the customer is at the table, so arriving and being seated are preconditions of UC-01, not use cases. Reservations, take-out and delivery are out of scope.

2. Actors

Actor	Type	Role in the scenario
Customer	Primary (person)	The only primary actor. The customer starts every use case: ordering, eating and paying for the meal, and optionally ordering beer.
Server	Secondary (person)	Takes and confirms the order, checks ID when beer is ordered, serves the food and presents the bill.
Cook	Secondary (person)	Prepares the food the customer ordered.
Bartender	Secondary (person)	Pours the beer once the order has passed the age check. In Ontario, anyone who serves alcohol must hold Smart Serve certification.
Card Payment Processor	Secondary (system)	The bank or card network that approves debit and credit payments. It sits outside the restaurant, so it is shown as a «system» actor.

3. Use case catalogue and relationship rationale

ID	Use case	Type	Relationship	Why this relationship
UC-01	**Order Food**	Base	Includes UC-04 Extended by UC-05	Customer goal 1 in the scenario.
UC-02	**Eat Food**	Base	Includes UC-07	Customer goal 2 in the scenario.
UC-03	**Pay for Food**	Base	Includes UC-08 Extended by UC-09, UC-10	Customer goal 3 in the scenario.
UC-04	Prepare Food	Included	UC-01 **«include»** UC-04	An order is not fulfilled until the kitchen prepares it, so every order includes this step.
UC-05	Order Beer	Extending	UC-05 **«extend»** UC-01	The customer only *might* order beer, and Order Food is complete without it. Condition: the customer wants beer.
UC-06	Verify Legal Drinking Age	Included	UC-05 **«include»** UC-06	Required every time beer is ordered: no alcohol for anyone under 19 in Ontario (BR-01).
UC-07	Serve Food	Included	UC-02 **«include»** UC-07	The food must reach the table before it can be eaten.
UC-08	Present Bill	Included	UC-03 **«include»** UC-08	The customer cannot pay without an itemized bill for what was ordered.
UC-09	Process Card Payment	Extending	UC-09 **«extend»** UC-03	Needed only for debit or credit; cash payment skips it. Condition: the customer pays by card.
UC-10	Add Tip	Extending	UC-10 **«extend»** UC-03	Tipping is optional. Condition: the customer chooses to tip.

4. Modelling decisions

- **One primary actor.** Only the Customer starts a use case. Staff and the payment processor only support the customer's goals, so they are secondary actors on the opposite side of the boundary.
- **Arrow direction.** **«include»** points from the base to the step it always performs. **«extend»** points from the optional use case back to its base, which is complete without it.
- **No extra actor lines.** Included and extending use cases run inside a base use case, so its actors already take part (for example, the Server checks ID within Order Food).
- **Business rules kept separate.** The age rule is recorded once as BR-01 and referenced from UC-06, as the BABOK recommends.
- **Naming and detail.** Every use case is named with a verb and a noun. Staff tasks such as plating food are left for next module's BPMN process models.

Use Case Specification: UC-01 Order Food

Completed using the Module 8 Use Case Template | Adekunle Adeniji

Use Case Name	**UC-01 Order Food**		
Date Created	November 3, 2026	**Last Updated**	November 3, 2026
Goal	The customer has ordered the meal they want, the order is recorded correctly, and the kitchen has prepared it, ready to be served.		
Actors	**Primary:** Customer. **Secondary:** Server (takes and confirms the order, checks ID for beer), Cook (prepares the food), Bartender (pours beer, only when UC-05 Order Beer extends this use case).		
Preconditions	The restaurant is open and the kitchen is operating. The customer is seated and has a menu. The point-of-sale (POS) ordering system is available.		
Trigger	The customer signals that they are ready to order, for example by closing the menu or calling the server over.		
Flow of Events	**Basic flow** 1. The Customer signals that they are ready to order. 2. The Server comes to the table and asks for the order. 3. The Customer chooses menu items and states any preferences or allergies. [Extension point: Beverage Order] 4. The Server reads the order back and the Customer confirms it. 5. The Server enters the order into the POS, which sends it to the kitchen and adds the items to the Customer's bill. 6. **Include UC-04 Prepare Food:** the Cook prepares the items as ordered. 7. The prepared food is ready for UC-07 Serve Food. The use case ends. **Alternative flows** **A1 Order Beer** (**UC-05 extends** at step 3). Condition: the Customer asks for beer. A1.1 **Include UC-06 Verify Legal Drinking Age:** the Server checks the Customer's government photo ID. A1.2 The Customer is 19 or older, so the beer is added to the order and sent to the bar, and the Bartender pours it. A1.3 The flow resumes at step 4. **A2 Item unavailable** (at step 5). The POS shows that an item is sold out. The Server tells the Customer, who chooses a substitute. The flow resumes at step 4. **A3 Allergy declared** (at step 3). The Server flags the allergy on the order so the Cook follows the allergen procedure. The flow resumes at step 4. **Exception flows** **E1 Age not verified** (at A1.1). The Customer has no valid ID or is under 19. The beer is refused, and the food order continues at step 4 without beer. **E2 Customer leaves** (before step 4). No order is sent to the kitchen and nothing is added to the bill. The use case ends without success.		
Post-conditions	**Success:** The confirmed order, plus any beer, is on the Customer's bill, and the kitchen has prepared the food, ready for UC-07 Serve Food. **Failure:** No order reaches the kitchen and the bill has no items.		
Other	**Related use cases:** includes UC-04 Prepare Food; extended by UC-05 Order Beer, which includes UC-06 Verify Legal Drinking Age. UC-01 is followed by UC-02 Eat Food and UC-03 Pay for Food. **Business rule:** BR-01: alcohol is served only to customers aged 19 or older (Ontario). Staff ask for government photo ID whenever a customer's age is in doubt. **Data inputs:** menu selections, preferences and allergies, and ID (beer only). **Outputs:** kitchen order ticket, bar order, and line items on the bill. **Departments:** Front of house (Server, Bartender); Back of house (Cook).		